A smart, task-oriented conversation engine prototype

Presented by **Ali NC**., a data enthusiast

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★ System to answer/guide user at each turn t, so that his requests w.r.t set of pre-defined functional evidences (raised from t = 1 until t = T) are all fulfilled (by t = T).

Cf. VCA domain

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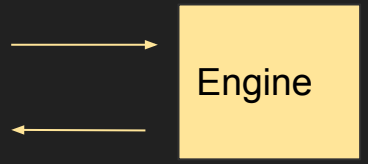

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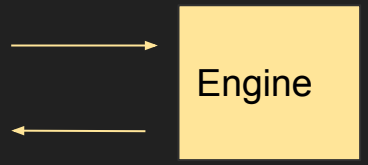

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BOT: “ring identified success. Size details: 30.5 mm, Stone details: made from Malachite, Engraving: true: price_spec_leadtime. Here is history of vca: … . End! Want something else, feel free please ?

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1	unnamed	5	5	0	0	30.5
2	HiJewel	7	2	2	4	undef
3	Exoticus	2	1	3	4	30.5
4	Ali Ncibi	7	4	0	1	15
5	Emilie-bennad's	6	4	0	2	undef
6	Sunsea small model	5	5	0	3	20
7	unnamed	3	2	3	0	20
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	seid	name	parent	is_endpoint
0	1	Know VCA history	undefined	true
1	2	Contact ambassador	undefined	true
2	3	Access the FAQ	undefined	false
3	4	Purchase question	3	false
4	5	Warranty question	4	true

Extends to any closed ontological domain

nothing special to vca functional ontological domain

FINITE TYPES OF EVIDENCES
USER MIGHT INVOKE IN
ANY CLOSED DOMAIN

nothing special to vca functional ontological domain

FINITE TYPES OF EVIDENCES USER MIGHT INVOKE IN ANY CLOSED DOMAIN

I) central-object internal attribute evidences

want to get: attr X, attr Y, etc. of my object ..

II) central-object external actions evidences

want to get: action a, action b, etc. about my object ..

III) central-object filtering evidences

an object with x val for X attr, y val for Y attr, etc.

IV) object-free evidences

I have a misc quest. a, misc quest. b, etc.

nothing special to vca functional ontological domain

FINITE TYPES OF EVIDENCES USER MIGHT INVOKE IN ANY CLOSED DOMAIN

central-object:
high end brand product
(dior, valentino, ..)

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Engine Workflow

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cf. VCA domain

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STEP ONE

Training an automatic (not regex, AI based) text tagger to understand domain evidences in user utterances

Engine Workflow > cf. VCA domain

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Domain evidences → Domain slots

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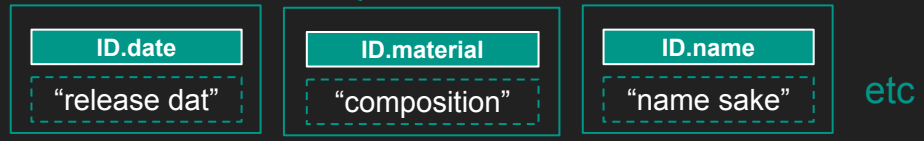

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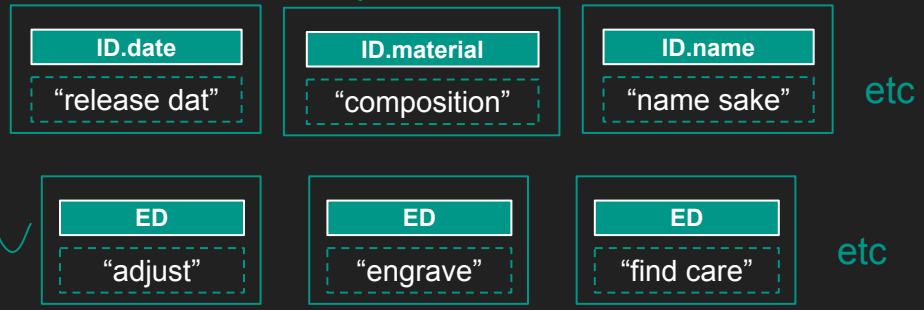

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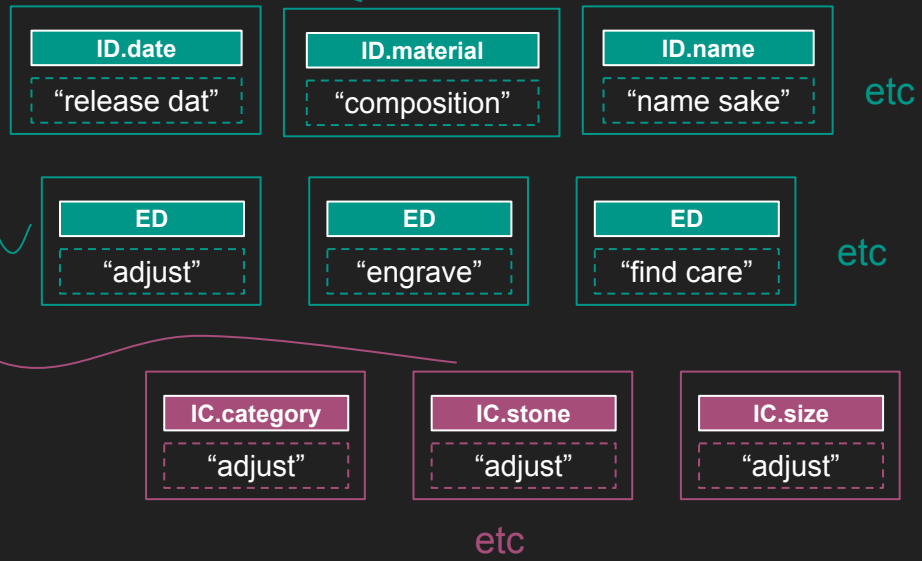

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ID.date
"release dat"

ID.material
"composition"

ID.name
"name sake"

etc

ED
"adjust"

ED
"engrave"

ED
"find care"

etc

PSD
"access faq"

PSD
"history of vca"

etc

IC.category
"bracelet"

IC.stone
"onyx"

IC.size
"larger than 20 mm"

etc

ED
"etch"

ID.name
"name"

ID.category
"kind"

ID.clasps
"clasp type"

ID.stone
"gem"

ID.material
"compos"

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ID.collection
"collection name"

ID.size
"length"

ID.date
"release date"

IC.name
"Flora small"

IC.category
"Bracelet"

IC.clasps
"Blade"

IC.material
"rose gold"

IC.stone
"malachite"

IC.collection
"Zodiac"

PSD
"acccess faq"

IC.size
"larger than 20 mm"

IC.date
"before 13/05.."

1 slot to capture user requests % product external-details

ED
"etch"

ID.name
"name"

ID.category
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ID.clasps
"clasp type"

ID.stone
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ID.material
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8 slots to capture user requests % product internal-details

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8 slots to capture user internal-constraints on the requested product

PSD
"acccess faq"

1 slot to capture user requests % product-free details (pure service details)

Engine Workflow > cf. VCA domain

STEP ONE: Training an automatic (not regex, AI based) text tagger to understand domain evidences in user utterances

Domain slots → Supervised learning corpus

Embed domain slots into local, then global labeled synthetic sentences.

ID.stone

"X"

X in ["stone", "gem stone", gemstone" , "gestone", "pierre", ..]

list of : raw value for the slot
+ pre-defined potential Synonyms*
(real synos, typos, transpositions,, lower/uppper-case ..) for that value

"get {ID.stone_placeholder} details"
"get {ID.stone_placeholder} information"
"show {ID.stone_placeholder} details"
"show {ID.stone_placeholder} information"
…

list of : pre-defined possible generic slot-specific local patterns ⇔ generic local contexts

"get stone details"
"get stone information"
"show stone details"
"show stone information"
…

"get pierre details"
"get pierre information"
"show pierre details"
"show pierre information"
…

synthetic samples of local slot-specific labeled strings

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ID.size

"X"

X in ["size", "length", "width", ..]

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for that value

"get {ID.stone_placeholder} details"
"get {ID.stone_placeholder} information"
"show {ID.stone_placeholder} details"
"show {ID.stone_placeholder} information"
…

list of : pre-defined possible generic
slot-specific local patterns ⇔
generic local contexts

"get size details"
"get size information"
"show size details"
"show size information"
…

"get width details"
"get width information"
"show width details"
"show width information"
…

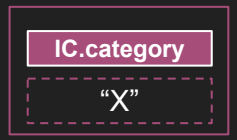
synthetic
samples of local
slot-specific
labeled strings

Engine Workflow > cf. VCA domain

STEP ONE: Training an automatic (not regex, AI based) text tagger to understand domain evidences in user utterances

Domain slots → Supervised learning corpus

Embed domain slots into local, then global labeled synthetic sentences.



"a {IC.category_placeholder} product"
"a {IC.category_placeholder} jewel"
"of kind {IC.category_placeholder}"
"of kind {IC.category_placeholder}"
…

"a bracelet product"
"a bracelet jewel"
"of kind bracelet"
"of kind bracelet"
…

"a bangle product"
"a bangle jewel"
"of kind bangle"
"of kind bangle"
…

synthetic samples of local slot-specific labeled strings

X in ["bracelet", "arm band", "bangle" , ..]

list of : raw value for the slot + pre-defined potential Synonyms* (real synos, typos, transpositions ..) for that value

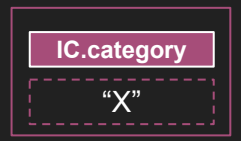
list of : pre-defined possible generic slot-specific local patterns ⇔ generic local contexts

Engine Workflow > cf. VCA domain

STEP ONE: Training an automatic (not regex, AI based) text tagger to understand domain evidences in user utterances

Domain slots → Supervised learning corpus

Embed domain slots into local, then global labeled synthetic sentences.



"a {IC.category_placeholder} product"
"a {IC.category_placeholder} jewel"
"of kind {IC.category_placeholder}"
"of kind {IC.category_placeholder}"
…

"a watch product"
"a watch jewel"
"of kind watch"
"of kind watch"
…

"a timepiece product"
"a timepiece jewel"
"of kind timepiece"
"of kind timepiece"
…

synthetic samples of local slot-specific labeled strings

X in ["watch", "clock", "timepiece" , ..]

list of : raw value for the slot + pre-defined potential Synonyms* (real synos, typos, transpositions ..) for that value

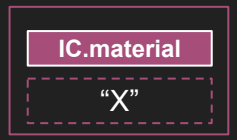
list of : pre-defined possible generic slot-specific local patterns ⇔ generic local contexts

Engine Workflow > cf. VCA domain

STEP ONE: Training an automatic (not regex, AI based) text tagger to understand domain evidences in user utterances

Domain slots → Supervised learning corpus

Embed domain slots into local, then global labeled synthetic sentences.



"made of {IC.material_placeholder}"
"made from {IC.material_placeholder}"
"contains {IC.material_placeholder} material"
"composed of {IC.material_placeholder} substance"
…

X in ["White gold", "white gold", "white-gold" , ..]

list of : pre-defined possible generic slot-specific local patterns ⇔ generic local contexts

"made of White gold"
"made from White gold"
"contains White gold material"
"composed of White gold substance"
…

synthetic samples of local slot-specific labeled strings

"made of white-gold"
"made from white-gold"
"contains white-gold material"
"composed of white-gold substance"
…

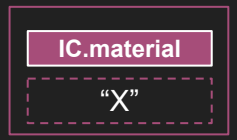
list of : raw value for the slot + pre-defined potential Synonyms* (real synos, typos, transpositions ..) for that value

Engine Workflow > cf. VCA domain

STEP ONE: Training an automatic (not regex, AI based) text tagger to understand domain evidences in user utterances

Domain slots → Supervised learning corpus

Embed domain slots into local, then global labeled synthetic sentences.



"made of {IC.material_placeholder}"
"made from {IC.material_placeholder}"
"contains {IC.material_placeholder} material"
"composed of {IC.material_placeholder} substance"
…

X in ["Rose gold", "rose gold", "rose-gold" , ..]

list of : pre-defined possible generic slot-specific local patterns ⇔ generic local contexts

"made of Rose gold"
"made from Rose gold"
"contains Rose gold material"
"composed of Rose gold substance"
…

synthetic samples of local slot-specific labeled strings

"made of rose-gold"
"made from rose-gold"
"contains rose-gold material"
"composed of rose-gold substance"
…

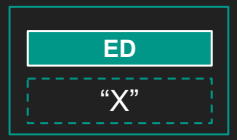
list of : raw value for the slot + pre-defined potential Synonyms* (real synos, typos, transpositions ..) for that value

Engine Workflow > cf. VCA domain

STEP ONE: Training an automatic (not regex, AI based) text tagger to understand domain evidences in user utterances

Domain slots → Supervised learning corpus

Embed domain slots into local, then global labeled synthetic sentences.



"perform {ED_placeholder} service"
"do {ED_placeholder} action"
"get {ED_placeholder} service details"
"get {IDED_placeholder} action information"
…

X in ["Adjust", "adjust", "adjustment", "resize", ..]

list of : pre-defined possible generic slot-specific local patterns ⇔ generic local contexts

"perform Adjust service"
"do Adjust action"
"get Adjust service details"
"get Adjust action information"
…

"perform resize service"
"do resize action"
"get resize service details"
"get resize action information"
…

synthetic samples of local slot-specific labeled strings

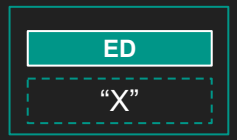
list of : raw value for the slot + pre-defined potential Synonyms* (real synos, typos, transpositions,, lower/uppper-case ..) for that value

Engine Workflow > cf. VCA domain

STEP ONE: Training an automatic (not regex, AI based) text tagger to understand domain evidences in user utterances

Domain slots → Supervised learning corpus

Embed domain slots into local, then global labeled synthetic sentences.



X in ["Engrave", "engrave" , "etch", "add text", "add drawing" ..]

list of : raw value for the slot
+ pre-defined potential Synonyms*
(real synos, typos, transpositions,, lower/uppper-case ..) for that value

"perform {ED_placeholder} service"
"do {ED_placeholder} action"
"get {ED_placeholder} service details"
"get {IDED_placeholder} action information"
…

list of : pre-defined possible generic slot-specific local patterns ⇔ generic local contexts

"perform Engrave service"
"do Engrave action"
"get Engrave service details"
"get Engrave action information"
…

"perform add drawing service"
"do add drawing action"
"get add drawing service details"
"get add drawing action information"
…

synthetic samples of local slot-specific labeled strings

Engine Workflow > cf. VCA domain

STEP ONE: Training an automatic (not regex, AI based) text tagger to understand domain evidences in user utterances

Domain slots → Supervised learning corpus

Embed domain slots into local, then global labeled synthetic sentences.

"get into {PSD_placeholder} service"
"show me {PSD_placeholder} service"
…

X in ["FAQ", "access faq", "Access faq" , "frequently asked", "faq" ..]

list of : pre-defined possible generic slot-specific local patterns ⇔ generic local contexts

"get into FAQ service"
"show me FAQ service"
…

"get into faq service"
"show me faq service"
…

synthetic samples of local slot-specific labeled strings

list of : raw value for the slot + pre-defined potential Synonyms* (real synos, typos, transpositions,, lower/uppper-case ..) for that value

Engine Workflow > cf. VCA domain

STEP ONE: Training an automatic (not regex, AI based) text tagger to understand domain evidences in user utterances

Domain slots → Supervised learning corpus

Embed domain slots into local, then global labeled synthetic sentences.

"get into {PSD_placeholder} service"
"show me {PSD_placeholder} service"
…

X in ["Warranty", "warranty", "grant" ,.]

list of : pre-defined possible generic slot-specific local patterns ⇔ generic local contexts

"get into Warranty service"
"show me Warranty service"
…

"get into grant service"
"show me grant service"
…

synthetic samples of local slot-specific labeled strings

list of : raw value for the slot + pre-defined potential Synonyms* (real synos, typos, transpositions,, lower/uppper-case ..) for that value

Engine Workflow > cf. VCA domain

STEP ONE: Training an automatic (not regex, AI based) text tagger to understand domain evidences in user utterances

Domain slots → Supervised learning corpus

Embed domain slots into local, then global labeled synthetic sentences.

"get pierre details"
"get pierre information"
"show pierre details"
"show pierre information"
…

"a bangle product"
"a bangle jewel"
"of kind bangle"
"of kind bangle"
…

"get into FAQ service"
"show me FAQ service"
…

"perform resize service"
"do resize action"
"get resize service details"
"get resize action information"
…

"made of Rose gold"
"made from Rose gold"
"contains Rose gold material"
"composed of Rose gold substance"
…

Engine Workflow > cf. VCA domain

STEP ONE: Training an automatic (not regex, AI based) text tagger to understand domain evidences in user utterances

Domain slots → Supervised learning corpus

Embed domain slots into local, then global labeled synthetic sentences.

Concatenate random samples of local contexts for different distinct slots

Engine Workflow > cf. VCA domain

STEP ONE: Training an automatic (not regex, AI based) text tagger to understand domain evidences in user utterances

Domain slots → Supervised learning corpus

Embed domain slots into local, then global labeled synthetic sentences.

“get pierre details” + [OF] +“my bangle product” + “made of rose-gold” + [ALSO] + get into FAQ service

…..

…..

“get size details” + [OF] +“my timepiece jewel” + “composed from White gold” + [ALSO] + access faq service

…..

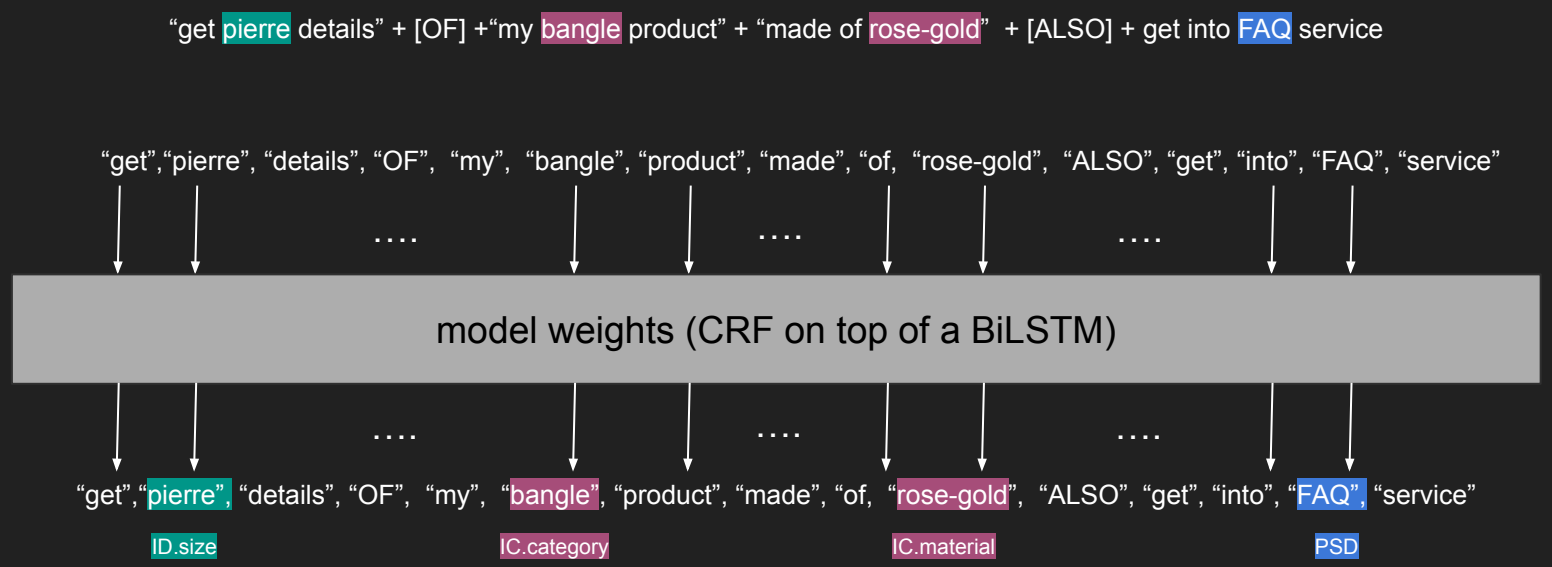
…..

Engine Workflow > cf. VCA domain

STEP ONE: Training an automatic (not regex, AI based) text tagger to understand domain evidences in user utterances

Supervised learning corpus → Learn a Sequence tagging model

Use global synthetic sentences alike generated to learn a sequence tagging model (CRF, BiLSTM, BiLSTM-CRF, …)



Engine Workflow

cf. VCA domain

STEP TWO

Dialog management (achieved with the help of a pre-trained text tagger on the domain-specific corpus)

Engine Workflow > cf. VCA domain

STEP TWO: Dialog management (achieved with the help of a pre-trained text tagger on the domain-specific corpus

t = 0: Dialog state (t = 0) := { }

t = 1 :
User : “I want to get sizzze details and fuel type of my product”

pre-trained* tagging module (CRF, BiLSTM, BiLSTM-CRF, xFormer, etc.)

I want to get sizzze details and fuel type of my product”

Reviewer/cleaner

map: (pre-defined)
raw slot value as found in db → its potential synonyms

Size	[size, siz, length, width, …]
…	..
..	..

sizzze

fuel type

{Size}

0 <= (encoder based, Jaro-Winkler, other fuzzy string matching techniques .) distance ([synonyms], entity) <= 1
(consider as match asap similarity = 1 - distance > some threshold (eg. 90%)

Engine Workflow > cf. VCA domain

STEP TWO: Dialog management (achieved with the help of a pre-trained text tagger on the domain-specific corpus

t = 0: Dialog state (t = 0) := { }

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I want to get sizzze details and fuel type of my product”

Reviewer/cleaner

sizzze

fuel type

{Size}

+

map: (pre-defined)
raw slot value as found in db → its potential synonyms

Size	[size, siz, length, width, …]
…	..
..	..

0 <= (encoder based, Jaro-Winkler, other fuzzy string matching techniques .) distance ([synonyms], entity) <= 1
(consider as match asap similarity = 1 - distance > some threshold (eg. 90%)

Engine Workflow > cf. VCA domain

STEP TWO: Dialog management (achieved with the help of a pre-trained text tagger on the domain-specific corpus

t = 0: Dialog state (t = 0) := { }

t = 1 :
User : “I want to get sizzze details and fuel type of my product”

pre-trained* tagging module (CRF, BiLSTM, BiLSTM-CRF, xFormer, etc.)

I want to get sizzze details and fuel type of my product”

Reviewer/cleaner

{Size}

+

Dialog state (t = 1) = CUREENT + Dialog state (t = 0)

map: (pre-defined)
raw slot value as found in db → its potential synonyms

Size	[size, siz, length, width, …]
…	..
..	..

sizzze

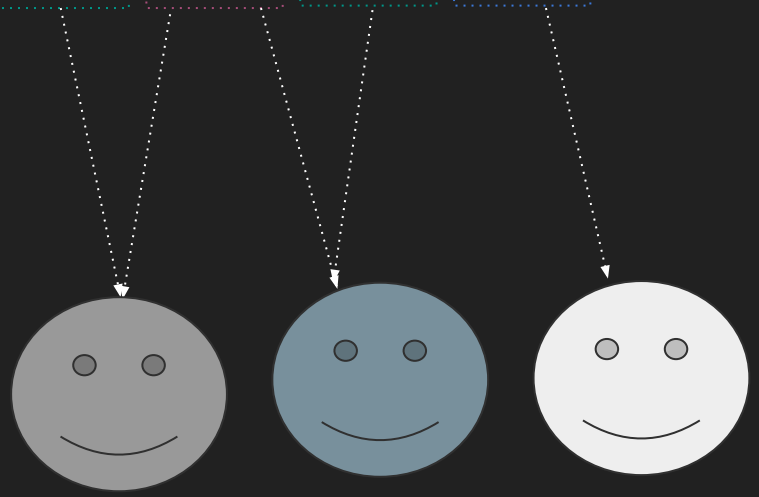
fuel type

0 <= (encoder based, Jaro-Winkler, other fuzzy string matching techniques .) distance ([synonyms], entity) <= 1
(consider as match asap similarity = 1 - distance > some threshold (eg. 90%)

Engine Workflow > cf. VCA domain

STEP TWO: Dialog management (achieved with the help of a pre-trained text tagger on the domain-specific corpus

t = 1: Dialog state (t = 1) := { Size }



Engine Workflow > cf. VCA domain

STEP TWO: Dialog management (achieved with the help of a pre-trained text tagger on the domain-specific corpus

t = 1: Dialog state (t = 1) := { Size }

Handle questions about product attributes

Handle questions about product services

Handle product-free questions

Engine Workflow > cf. VCA domain

STEP TWO: Dialog management (achieved with the help of a pre-trained text tagger on the domain-specific corpus

t = 1: Dialog state (t = 1) := { Size }

Handle questions about product attributes

Handle questions about product services

Handle product-free questions

Engine Workflow > cf. VCA domain

STEP TWO: Dialog management (achieved with the help of a pre-trained text tagger on the domain-specific corpus

t = 1: Dialog state (t = 1) := { Size }

Handle questions about product attributes

Handle questions about product services

Handle product-free questions

Engine: “didn’t get enough info to answer -> ask for internal constraints to be able to isolate a product”

Engine Workflow > cf. VCA domain

STEP TWO: Dialog management (achieved with the help of a pre-trained text tagger on the domain-specific corpus

t = 1: Dialog state (t = 1) := {Size }

t = 2 :
User : “made from white gold, I also want to etch it and access to history of vca”

pre-trained* tagging module (CRF, BiLSTM, BiLSTM-CRF, xFormer, etc.)

made from white gold, I also want to etch it and access to history of vca

Reviewer/cleaner

{white gold, history of vca,etch }

map: (pre-defined)
raw slot value as found in db → its potential synonyms

White gold	[size, siz, length, width, …]
Engrave	[engrave, etch, etching, drawing ,..]
VCA Hist.	[story of vca, vca history, history of vca ..]

white gold

etch

history of vca

0 <= (encoder based, Jaro-Winkler, other fuzzy string matching techniques .) distance ([synonyms], entity) <= 1
(consider as match asap similarity = 1 - distance > some threshold (eg. 90%)

Engine Workflow > cf. VCA domain

STEP TWO: Dialog management (achieved with the help of a pre-trained text tagger on the domain-specific corpus

t = 1: Dialog state (t = 1) := {Size }

t = 2 :
User : “made from white gold, I also want to etch it and know about vca’s history”

pre-trained* tagging module (CRF, BiLSTM, BiLSTM-CRF, xFormer, etc.)

made from white gold, I also want to etch it and know about Vca’s history

Reviewer/cleaner

{white gold, history of vca,etch }

+

Dialog state (t = 2) = CUREENT + Dialog state (t = 1)

map: (pre-defined)
raw slot value as found in db → its potential synonyms

White gold	[size, siz, length, width, …]
Engrave	[engrave, etch, etching, drawing ,..]
VCA Hist.	[story of vca, vca history, history of vca ..]

white gold

etch

Vca’s history

0 <= (encoder based, Jaro-Winkler, other fuzzy string matching techniques .) distance ([synonyms], entity) <= 1
(consider as match asap similarity = 1 - distance > some threshold (eg. 90%)

Engine Workflow > cf. VCA domain

STEP TWO: Dialog management (achieved with the help of a pre-trained text tagger on the domain-specific corpus

t = 1: Dialog state (t = 2) := { Size | white gold | etch | history of vca }

Handle questions about product attributes

Handle questions about product services

Handle product-free questions

Engine Workflow > cf. VCA domain

STEP TWO: Dialog management (achieved with the help of a pre-trained text tagger on the domain-specific corpus

t = 1: Dialog state (t = 2) := { Size | white gold | etch | history of vca }

Engine: “Tada! here are PSD “hist. of vca” details ..”

Handle questions about product attributes

Handle product-free questions

Engine: “Tada! product identified, here are size details ..”

Handle questions about product services

Engine: “Tada! product identified, here are etching details ..”

Time To Play

cf. VCA domain

Playing with a demo of the previous proto. up at:
https://share.streamlit.io/ylaxor/chat-bot-proto/app.py

Continuous development

- Augment synthetic training corpus continuously with more natural wordings + new synonyms (text paraphrasing methods, new real synonyms, ..)

- Track failing user utterances (where the tagger is unable to tag important words) continuously, analyze & debug them, and add them to training corpus to overcome them in next iteration.

- Review policy managers's policies (it's flexible, we can impose whatever policy we want, but we'd like it to be optimal ⇔ finish the conversation (fulfill requests) as soon as possible!

Limit performances

- Depends highly on training corpus quality/volume
- the tagger is a classifier, slots names are the classes → class imbalance problems in training data (control using f-beta metrics to assess classification results; Recall, Precision, harmonic average ..)
- Beware to sampling step while synthetic-data creation (keep the real feel, include as much as possible, be sure to not exclude important slots, beware to overfitting)
- Suggestion: train on synthetic while evaluating on well-chosen test corpus with more natural sentences

Extensions

- Extends to any similar closed ontological functional domain
  → only step one (preparing needed slots, their wordings, and a tagger to understand them in future utterances) needs to be readjusted!

- Extends instantly to any new languages
  → Just translate training corpus tokens at step one using DeepL, G-translate, or even our internal model

- Can accept multi-modal input data:
  → Audio → speechrecog. AI → text → we're done
  → Image → objects descriptor → text → we're done

Merci de votre attention